Mississippi Valley State University

CS 341 — Discrete Structures

Practice Set 37

Graphs, Degree, and the First Theorem

Covers Chartrand & Zhang, Section 12.1

Status of this set

Ungraded practice. Solutions released with it. **Draw each graph** before answering — a sketch catches most errors.

Part I: How to Practice This Material

The one tool

$$\sum_{v \in V(G)} \deg v = 2m$$

Everything in this set follows from it. Two consequences:

- The degree sum is always **even**.
- The number of **odd** vertices is always **even**.

Three mistakes to expect

1. **Forgetting the factor of 2.** The sum is $2m$, not m.
2. **Accepting an odd degree sum.** No such graph exists.
3. **Assuming a legal degree sum means the graph exists.** It is necessary, not sufficient.

Part II: Practice Problems

Section A — Reading a graph

1. Let $V = \{v_1, \ldots, v_6\}$ and $E = \{v_1v_2, v_1v_3, v_2v_3, v_3v_4, v_4v_5\}$. Draw the graph, then give its order, size, degree sequence, and the number of isolated vertices.

Section B — Using the degree sum

2. A graph has size 21. It has two vertices of degree 5, three of degree 6, and all remaining vertices have degree 2. What is its order?

3. For each, decide whether a graph with the stated degrees exists. Justify.

(a) $2, 2, 3, 3, 4$

(b) $1, 1, 2, 3, 3, 4$

(c) $1, 2, 2, 3, 3, 4$

(d) A 4-regular graph of order 7

Section C — Regular graphs

4. For which orders n does a 3-regular graph exist? Justify.

Section D — Writing

5. In four to six complete sentences, prove the First Theorem of Graph Theory and explain why it is a counting argument rather than an algebraic one.

Section E — Challenge (optional)

6. Show that in any graph of order $n \geq 2$, some two vertices have the same degree.